

5. (a) The table shows some properties of three plastics.

Plastic	Properties		
polycarbonate	high impact strength	good heat resistance	transparent
bakelite	rigid	good chemical resistance	good thermal insulator
coated nylon	good electrical insulator	flexible	waterproof

Use only the information in the table to answer this part-question.

State which plastic is used for making each of the following items.

Choose the **two** main properties that make the plastic suitable.

Give the reasons why these properties are important. [6 QER]





(b)

Recycling Plastics

Of all the materials we recycle, plastic is the most complex. The problem is that some plastic packaging is made from several different types of plastic. Different types of plastic are identified by different symbols.

Type of plastic	Symbol	Ease of recycling	Uses
PET		easy	soft drink bottles, rigid food packaging
HDPE		easy	milk cartons, cleaning product bottles, yoghurt pots
PVC		difficult	window frames, guttering
LDPE		difficult	food bags, shopping bags
PP		easy	margarine tubs, microwave meal trays
PS		difficult	some yoghurt pots, takeaway meal trays
other		cannot be recycled	mixed plastic products e.g. salad bags, crisp bags

Waste plastic is collected in different ways.

- Some local authorities collect all their recycling in one bin
- Others ask households to separate their plastics from the rest of their recycling

Sometimes plastic waste cannot be recycled and has to be sent to landfill or incinerated instead. This happens for several reasons.

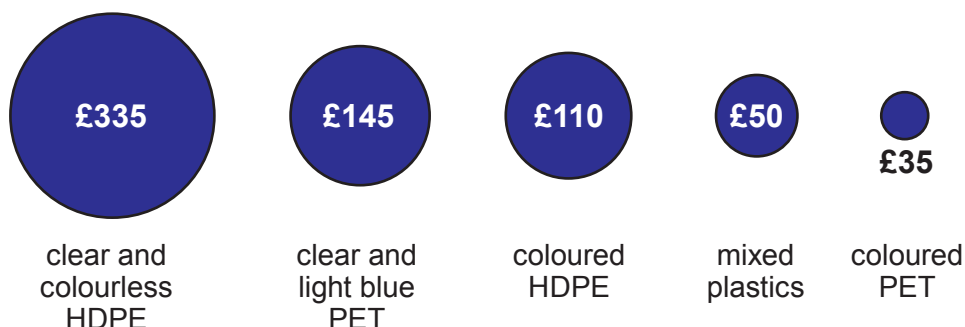
- People are confused about which plastics can and cannot be recycled
- The plastic packaging is contaminated with food waste
- In areas where recycling materials are collected in one bin, one type of waste can contaminate another

Most plastic can be recycled – but it is not always economical to do so.

- Bottles attract the best prices, especially clear ones
- Coloured plastic is less desirable; the colour cannot be removed so its use is limited
- Polystyrene is almost never recycled because there is no market for it



Maximum price per tonne of used bottles (2018)



- (i) Put a **tick (✓)** next to the correct statement.

[1]

all plastics are easy to recycle

☐

PET, HDPE and PP are easy to recycle

☐

PET, HDPE and PVC are difficult to recycle

☐

PVC, LDPE and PS cannot be recycled

☐

- (ii) Put a **tick (✓)** in **two** boxes which explain why some plastic items have to go to landfill.

[2]

the plastic items still contain food waste

☐

not enough plastic items have been collected to make recycling worthwhile

☐

the plastic item is made up of more than one type of plastic

☐

too many plastic items have been collected for recycling

☐

the plastic items are too large to fit in the incinerator

☐

- (iii) **Circle** the type of plastic for which there is the greatest demand for recycling.

[1]

coloured PET

clear and light blue PET

clear and colourless HDPE

coloured HDPE

